

4

ELECTRICAL POWER

This chapter introduces power, energy and work, starting with a review of mechanical energy and the various forms of energy. This is followed by an explanation of electrical power and how to calculate the power taken by an electrical load. We also look at the effect on power if the voltage or current changes, and how the efficiency of an electrical load affects the amount of power required to achieve a certain task.

Chapter outline

- 4.1 Energy and work
- 4.2 Power
- 4.3 Electrical power
- 4.4 Transposing the power equation
- 4.5 Power and Ohm's law
- 4.6 Power change with I, V or R changes



4.1 Energy and work

Electricity is a convenient form of energy that is obtained from other forms of energy. It is convenient because electrical energy can be produced near a source of energy, such as coal deposits, then transmitted over long distances to where it is needed. In effect, electricity is a means of transferring energy from one point to another. At the receiving end, electrical energy is converted to the required form, such as heat, mechanical, chemical or light, as outlined in Chapter 2. But what is energy?

Energy is the *ability to do work*. It can only exist in a form, such as heat or light, and energy in one form can disappear but the same amount of energy will appear in another form. That is, energy can only be *transformed* from one form to another, as it cannot be created or destroyed.

The unit of energy is the *joule*. When energy is transformed from one form to another, *work* is done. The amount of work done equals the amount of energy transformed, so work is also measured in joules. If no work is done, no energy is transformed. We can see this by looking at the most familiar form of energy: mechanical energy.

Mechanical energy

In 1687, Isaac Newton presented his three laws of motion, which laid the foundation for a study of classical mechanics. Newton's first law says: *An object at rest tends to stay at rest and an object in motion tends to stay in motion with the same speed and direction unless acted upon by an unbalanced (or external) force*. This law is often called 'the law of inertia'.

Pushing a trolley is an example, in which the force you apply overcomes the trolley's inertia and causes it to move against any external forces (e.g. friction or other forms of resistance) that are acting to stop the trolley. If there were no external opposing forces, once it is set in motion, the trolley would keep moving forever.

Another example is lifting a heavy object from ground level to a higher level. In both cases a force is required to move the object a certain distance and work is done (energy is transformed) in the process. That is, work is done when a force applied to an object causes it to move through a certain distance. Work is always against some resistance such as gravity, friction or other opposing forces.

work done = force $\times$ distance

where:

work done is measured in joules (J)

force is measured in newtons (N)

distance is measured in metres (m).

Joules



One joule of work is done when a force of one newton is moved through a distance of one metre, in the direction of the force.

It's important to understand that force has a direction, which makes it a *vector* quantity. (Vector quantities have direction, where scalar quantities have magnitude only. Examples of scalar quantities are time, speed, temperature, resistance.) Vectors (and phasors) are covered later in this book.

Force

Newton's second law says: *Acceleration is produced when a force acts on a mass*. The greater the mass of the object being accelerated, the greater the amount of force needed to accelerate the object. We all know that heavier objects require more force to move than lighter objects; however,

this law gives us an exact relationship between force, mass and acceleration. It can be expressed as a mathematical equation:

$$F = ma$$

where:

force (F) is in newtons

mass (m) is in kilograms

acceleration (a) is in metres per second per second (written as m/s^2).

Force is measured in newtons (N) where one newton is defined as the force required to accelerate a one kilogram mass by a rate of one metre per second per second (or m/s^2).

In many cases work is done against gravity, such as lifting an object. Acceleration in the above equation is, therefore, the acceleration due to gravity (g), which is 9.8 m/s^2 . Example 4.1 below explains this further.



Force

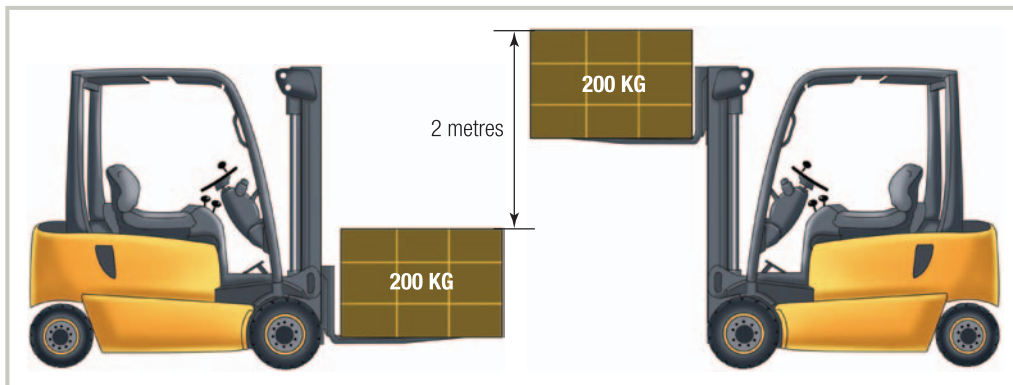


FIGURE 4.1

Electric forklift raising a 200 kg load through a distance of 2 metres

EXAMPLE 4.1

The electric forklift in Figure 4.1 is lifting a 200 kg load vertically through a distance of 2 metres. Find the force required, the work done and the amount of energy that has been transformed.

Solution

Values	$m = 200 \text{ kg}$ $d = 2 \text{ m}$ $a = \text{acceleration due to gravity} = 9.8 \text{ m/s}^2 \text{ (g)}$
Equation	$F = ma \text{ (or } mg)$ $= 200 \times 9.8$
Force	$(F) = 1960 \text{ N}$
Work	$= \text{force} \times \text{distance}$ $= 1960 \times 2$
Work	$= 3920 \text{ joules}$
Energy	$= \text{work done} = 3920 \text{ J}$
Answer	F = 1960 N, work done = energy converted = 3920 J = 3.92 kJ

The electrical energy used by the forklift in Example 4.1 is at least 3920 J, but as explained later, unless the forklift is 100 per cent efficient, additional energy is needed which will be expended, usually as heat (called losses).

Rotational energy

A familiar form of mechanical force is rotation, known as *torque*. Examples of torque are the force required to turn a steering wheel, or to tighten a nut with a spanner. Motors are generally rated by their torque, as well as their power. Figure 4.2(a) shows a torque wrench being used to tighten a bolt. It has a torque indicator which lets you tighten nuts and bolts to a value specified by the equipment manufacturer. (Bolted electrical connections are often given a torque specification.)

Torque is measured in newton-metres (Nm). As Figure 4.2(b) shows, the force to rotate a spanner might be applied anywhere along the length of the spanner. As you probably know, the further the distance (r) between the fulcrum and the applied force (F), the greater the torque (T). If the force is applied at right angles we can say that:

$$T = Fr$$

where:

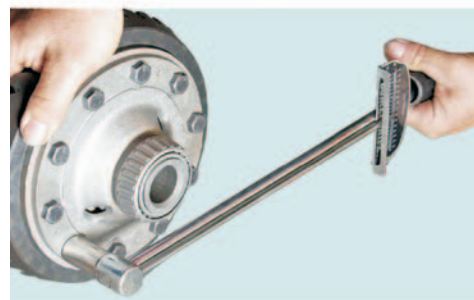
T = torque in newton-metres (Nm)

r = distance between fulcrum and applied force in metres (m)

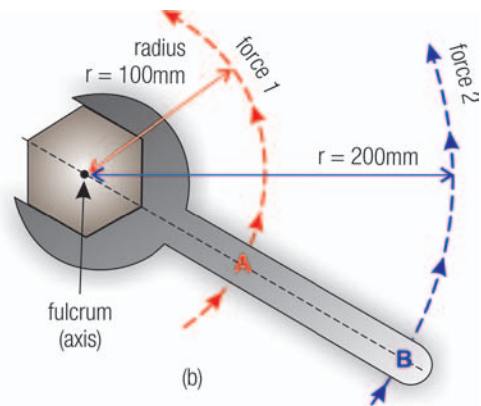
F = applied force in newtons (N).

FIGURE 4.2

(a) Using a torque wrench to tighten a bolt. (b) The torque (T) applied to the bolt depends on the applied force (F) and distance (r).



(a) Torque wrench with scale indicating the torque being applied to the bolt

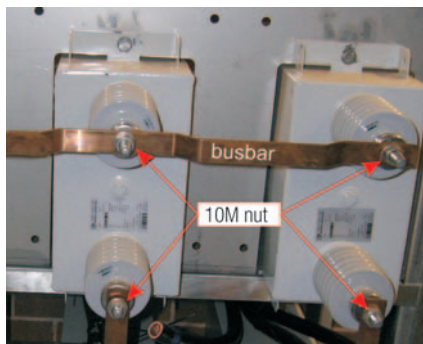


EXAMPLE 4.2

A manufacturer of electrical equipment specifies a torque of 40 Nm to correctly tighten the 10 M nuts in Figure 4.3. Calculate the force you need to apply to the torque wrench in Figure 4.2 (b) when (i) the force is applied at point A ($r = 100$ mm) and (ii) when the force is applied at point B ($r = 200$ mm).

FIGURE 4.3

Busbar attached with 10 mm bolts



Solution

Values $T = 40$ Nm
 $r =$ (i) 100 mm, (ii) 200 mm
 $F = ?$

Equation $T = r \times F$
 $F = \frac{T}{r}$ (by transposing the above equation)





(i) when $r = 100 \text{ mm} = 0.1 \text{ m}$

$$= \frac{40}{0.1}$$

Answer (i) = **400 N**

(ii) When $r = 200 \text{ mm} = 0.2 \text{ m}$

$$= \frac{40}{0.2}$$

Answer (ii) = **200 N**

Potential and kinetic energy

Mechanical energy can be either kinetic energy (energy of motion) or potential energy (stored energy of position). The kinetic energy of wind provides the energy to rotate a wind generator. A raised hammer has potential energy that is converted to kinetic energy when you cause the hammer to strike a nail.

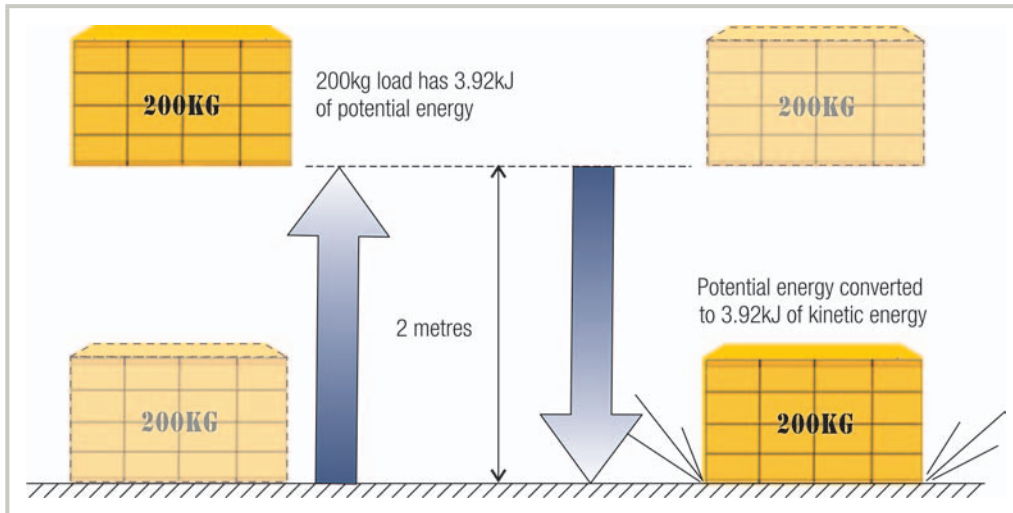


FIGURE 4.4

A raised, stationary mass has potential energy that is converted to kinetic energy when the mass is moving. When the moving mass is suddenly stopped, the kinetic energy is converted to other forms of energy.

For example, in Figure 4.4, a 200 kg load is raised through a distance of two metres. As calculated in Example 4.1, it takes 3.92 kJ of energy to do this. This energy is stored in the load, and if it is supported in some way, we can say the load has 3.92 kJ of potential energy. Newton's third law of motion says: *For every action there is an equal and opposite reaction*. The downward force exerted by the 200 kg load is counteracted by an upwards force from whatever is supporting the load.

If this force (support) is removed, the load will drop from its height of two metres so it strikes the ground. The 3.92 kJ of potential energy in the load will then be converted to 3.92 kJ of kinetic energy, which when the load hits the ground, is dissipated in other forms of energy, including sound and heat. The equation to find kinetic energy is:

$$\text{kinetic energy} = \frac{1}{2}mv^2 \text{ joules}$$

where:

m = mass in kilograms

v = velocity in metres per second (m/s).

Put into words, this equation says that the kinetic energy of an object equals half its mass multiplied by the square of its velocity. That is, if the velocity is doubled, the kinetic energy increases four times.

SUMMING UP...

- Energy is the ability to do work. It is measured in joules.
- Work is done when energy is transformed, also measured in joules.
- Force is a push or pull on an object that causes it to change shape or to move.
- In mechanics, work = force $\times$ distance, and one joule of work is done when a force of one newton is moved through a distance of one metre, in the direction of the force.
- Force = mass $\times$ acceleration (Newton's second law).
- Torque is a rotational force (vector quantity) and is measured in newton metres (Nm).

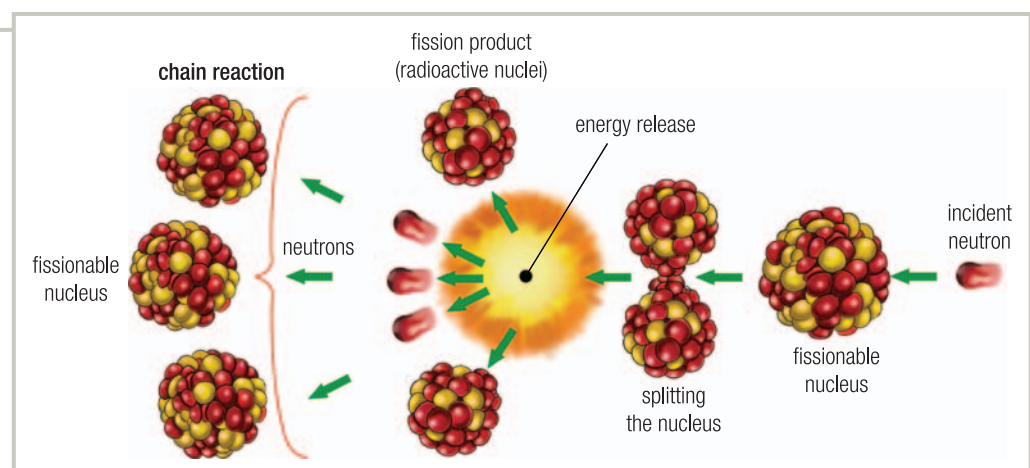
Forms of potential and kinetic energy

Energy has many forms, some of which are used to generate electricity. All forms of energy can be classified as either potential or kinetic, although potential energy is stored energy, waiting to be used. Examples of potential energy sources include:

- *Chemical energy* is the energy stored in the bonds of atoms and molecules. Examples are biomass, petroleum, natural gas and coal. When burned, the stored chemical energy is converted to thermal energy, which can be used to produce steam to drive a turbine. A battery has stored chemical energy, except it is converted directly to electrical energy when the battery is connected to a load.
- *Gravitational energy* is the energy stored as a result of an object's height (see Figure 4.4). The higher and heavier the object, the more gravitational energy is stored. Hydro-power is an example of gravitational energy, where a dam stores water that can be released to fall through a height and rotate a turbine. (The energy stored in a stretched rubber band or spring is called *elastic potential energy*.)
- *Nuclear energy* is energy stored in the nucleus of an atom – the energy that holds the nucleus together. Large amounts of energy are released when the nuclei are combined or split apart. Nuclear power plants split the nuclei of uranium atoms in a process called *fission* (see Figure 4.5) to produce the heat that generates steam. (The sun combines the nuclei of hydrogen atoms in a process called fusion.)

FIGURE 4.5

Nuclear power plants generate electricity by using heat obtained from a fission reaction, which releases large amounts of heat energy



Forms of kinetic energy used to generate electricity include:

- *Radiant energy* which is electromagnetic energy that travels in waves. Examples are visible light, x-rays and radio waves. Sunshine is radiant energy which can be converted to electricity using solar panels.
- *Thermal energy* (heat) is the vibration and movement of the atoms and molecules within substances. As an object is heated, its atoms and molecules move and collide more quickly. Geothermal energy is thermal energy in the Earth. Heat energy is produced from potential energy sources, or directly from the sun.
- *Motion energy* is energy in the movement of objects or substances. Examples are wind power from the kinetic energy of moving air, hydro-power from moving water, and power from rotational forces such as a steam turbine.

Sound is also kinetic energy, as it's the movement of waves through a substance (e.g. air). Sound energy is usually far less than other forms of energy. Electrical energy is the energy due to electrons flowing in a conductor, so it too is a form of kinetic energy.

4.2 Power

As already explained, when energy is transformed, work is done. However, we also need to consider how *quickly* energy is being transformed. In Figure 4.6, two men of equal weight are exercising during lunch. One man walks slowly and takes 20 minutes to walk a distance of one kilometre. The other is running and covers the same distance in five minutes. Who did the most work? Who used the most amount of energy?



FIGURE 4.6

If both men travel the same distance, who does the most work?

The answer is that the men have used the *same* amount of energy and have done the *same* amount of work. The difference is that the runner has taken less *time* to do so compared to his companion. The explanation is power.

Power is the *rate at which energy is used*, or the *rate at which work is done*. So while the walker is transforming less energy per second, he is doing it over a longer time than the runner. Therefore, they both transform the same amount of energy.

The unit of power is the *watt*, which has the letter **W** as its symbol. By definition, the power is *one watt if one joule of energy is transformed in one second*. Power is given the symbol **P**.

The watt is named after James Watt, an English inventor remembered mainly for his steam engines. Watt also introduced the term 'horsepower' to rate the power of his steam engines.

A horsepower is equal to 746 watts, and although this term is not a metric unit, there are still a lot of electric motors with a power rating given in horsepower (hp). The motor in a domestic refrigerator is usually rated at between $\frac{1}{4}$ hp to $\frac{1}{2}$ hp.

$$\text{power} = \frac{\text{work}}{\text{time}}$$

where:

power is in watts

work is in joules

time is in seconds.

EXAMPLE 4.3

The two electric forklifts in Figure 4.7 are lifting the same 200 kg load vertically through a distance of 2 metres. One does the lift in 2 seconds, the other takes 8 seconds. Find the power of both forklifts.

Solution

Values work = 3920 joules (from Example 4.1)
 time = (a) 2 seconds (b) 8 seconds
 power = ?

Equation $\text{power} = \frac{\text{work}}{\text{time}}$

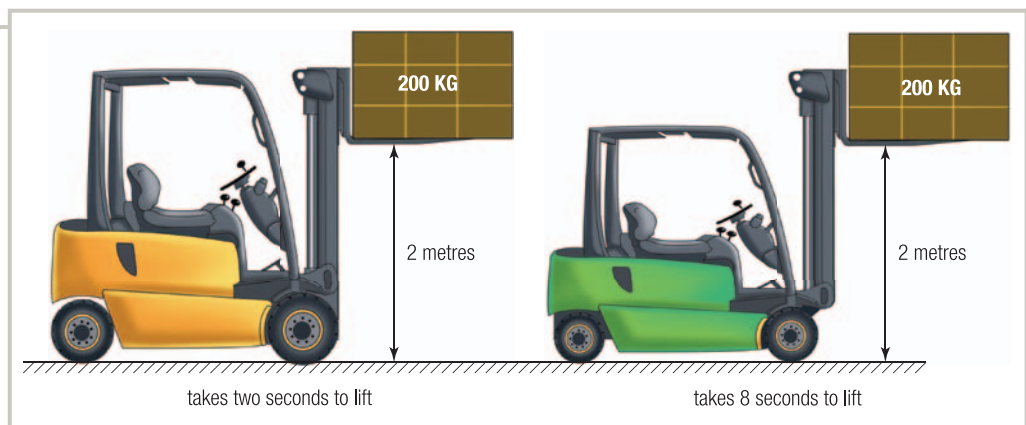
$$(a) \text{ power} = \frac{3920}{2} = 1960 \text{ watts}$$

$$(b) \text{ power} = \frac{3920}{8} = 490 \text{ watts}$$

Answer **(a) power = 1960 watts, (b) power = 490 watts**

FIGURE 4.7

The less power, the longer it takes to do the same work



Mechanical power

Mechanical power is a rating of how much work a machine such as a motor can do in a certain time. The mechanical power produced by a motor depends on its speed of rotation and the torque

it is delivering at that speed. Because we are dealing with a rotational quantity, we need to consider *angular velocity*, which is why the following equation includes the circular reference of pi (π).

$$P = 2\pi nT$$

where:

P = mechanical power in watts

2π is a circular reference equal to 6.28 (approx.)

n = revolutions per second

T = torque in newton-metres.

As motor speed is generally given in revolutions per minute (RPM), a more useful equation is:

$$P = \frac{2\pi NT}{60}$$

where: N = revolutions per minute (RPM).

EXAMPLE 4.4

Find the power (P) of a motor that is delivering 20 Nm of torque when it is running at 1440 RPM.

Solution

Values	torque (T) = 20 Nm RPM (N) = 1440 power (P) = ?
Equation	$P = \frac{2\pi NT}{60} = \frac{6.28 \times 20 \times 1440}{60}$
Answer	power = 3014 watts

Efficiency

The source of energy to power a motor has to come from another energy source. In Example 4.4, the motor is delivering slightly over 3 kW of mechanical power. This is called the output power. The input power is the power taken by the motor from its energy source. If there were no losses in the motor, it would require the same input power as its output power. That is, the motor would be 100 per cent efficient.

It is impossible for any machine to be 100 per cent efficient, as there are always losses in the process of converting energy from one form to another. In the case of an electric motor, the losses appear mainly as heat. If the motor was 100 per cent efficient, it would stay cool even when operated at full load. Efficiency is, therefore, a measure of power in and power out, and is sometimes given the symbol η (Greek letter *eta*). The general equation to find efficiency is:

$$\eta = \frac{\text{power out}}{\text{power in}} \times 100\%$$

where:

efficiency is expressed as a percentage (%)

power in and power out are expressed in watts.



Efficient use of energy helps towards sustainable energy

EXAMPLE 4.5

Find the efficiency of the motor in Example 4.4, if the power taken by the motor is 3500 watts.

Solution

Values power out = 3014 watts (as per Example 4.4)
 power in = 3500 watts
 efficiency (η) = ?

Equation $\eta = \frac{\text{power out}}{\text{power in}} \times 100\% = \frac{3014}{3500} \times 100\%$

Answer **efficiency = 86.1%**

SUMMING UP...

- All forms of energy can be classified as either potential or kinetic.
- Kinetic energy is energy in motion.
- Potential energy is energy in storage.
- Power is the rate of doing work. It has the symbol P and is measured in watts.
- Efficiency of a machine equals its power out divided by its power in, multiplied by 100.
- Efficiency gives a measure of the losses in a machine and is expressed as a percentage.

EXERCISE 4.1

- 1 Calculate the force required to lift a toolbox weighing 10 kg.
- 2 How much work is done if you lift the 10 kg toolbox through a vertical distance of 2.5 m?
- 3 How much potential energy is there in the 10 kg toolbox when it's resting 2.5 m from the ground?
- 4 If you drop the toolbox, what is its kinetic energy when it hits the ground?
- 5 When tightening a nut, if you apply a force of 300 N to a spanner at a point 150 mm from the centre of the nut, how much torque are you applying to the nut?
- 6 Find the kinetic energy of a 200 kg mass travelling at a velocity of 6 m/s.
- 7 How many watts does one horsepower represent?
- 8 How much power do you develop if you climb a 20 m long ladder in two minutes? Assume you weigh 60 kg and you are carrying an extra 5 kg of tools and equipment.
- 9 A car engine is delivering 220 Nm of torque at 1950 RPM. Calculate the power it is producing.
- 10 A machine requires 500 W of power and delivers 450 W of useful output power. Calculate the losses and the efficiency of the machine.

4.3 Electrical power

As already explained, the power is *one watt if one joule of energy is transformed in one second*. Electrical energy is used only when a voltage causes a current to flow (kinetic energy). Electrical power is the *rate* at which electrical energy is transformed into another form, such as heat.

Electrical power, like mechanical power, is measured in watts. One watt of electrical power is dissipated in a resistor when a voltage of one volt causes a current of one ampere to flow through the resistor. (By Ohm's law, the resistance of the circuit is one ohm.)

The general equation to find electrical power is:

power = voltage $\times$ current ($P = VI$)

where:

P = power in watts

V = voltage in volts

I = current in amperes.



Calculating
electrical power

EXAMPLE 4.6

Calculate the power being dissipated by the lamp in Figure 4.8 below.

Solution

Values $V = 24$ volts
 $I = 1.5$ amperes
 $P = ?$

Equation $P = VI = 24 \times 1.5$

Answer **$P = 36$ watts**

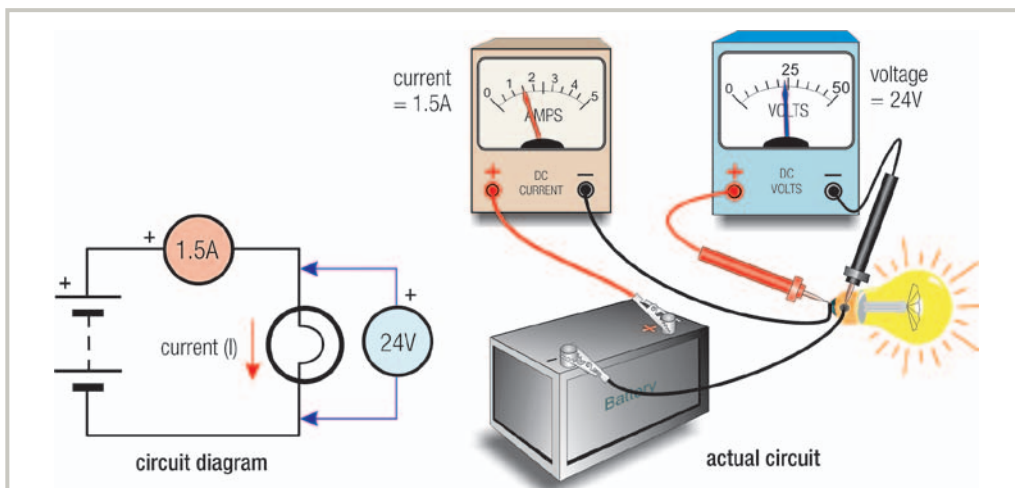


FIGURE 4.8

4.4 Transposing the power equation

As with Ohm's law, the power equation $P = VI$ can be transposed to make either voltage (V) or current (I) the subject. This gives equations that let you find the voltage (or current) if the power and the current (or voltage) are known.

To make voltage the subject

- 1 Write the equation: ($P = VI$).
- 2 Divide both sides by I : $\left(\frac{P}{I} = \frac{VI}{I}\right)$.
- 3 Cancel out the I 's on the right-hand side of the equation: $\left(\frac{P}{I} = V\right)$.
- 4 Rewrite the equation with V on the left-hand side giving: $V = \frac{P}{I}$.

To make current the subject

- 1 Write the equation: ($P = VI$)
- 2 Divide both sides by V : $\left(\frac{P}{V} = \frac{VI}{V}\right)$
- 3 Cancel out the V 's on the right-hand side of the equation: $\left(\frac{P}{V} = I\right)$
- 4 Rewrite the equation with I on the left-hand side giving: $I = \frac{P}{V}$.

This gives three equations that relate power, voltage and current:

$$P = VI$$

$$V = \frac{P}{I}$$

$$I = \frac{P}{V}$$

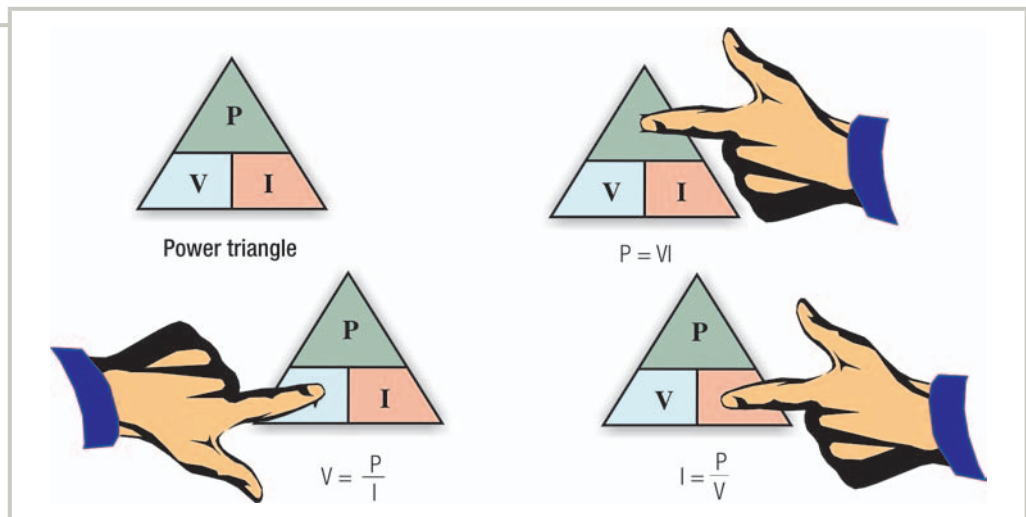
These equations, like those for Ohm's law, can be arranged in a triangle. As before, cover the term you want to find, and the equation is the other two terms (see Figure 4.9).

Equations
relating
power,
voltage and
current



FIGURE 4.9

Power triangle. Cover the term you want to find, and the equation is the other two terms



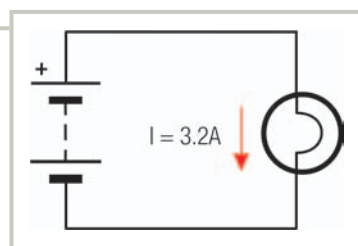
The next two examples show how to use these equations.

EXAMPLE 4.7

Find the voltage across the lamp in the circuit of Figure 4.10.

FIGURE 4.10

Circuit for Example 4.7



Solution

Values	$P = 40 \text{ watts}$ $I = 3.2 \text{ amperes}$ $V = ?$
Equation	$V = \frac{P}{I} = \frac{40}{3.2}$
Answer	$V = 12.5 \text{ volts}$

EXAMPLE 4.8

How much current does an electric heater rated at 750 W take when it's connected to a 230 V supply?

Solution

Values $P = 750$ watts
 $V = 230$ volts
 $I = ?$

Equation $I = \frac{P}{V} = \frac{750}{230}$

Answer **$I = 3.26$ amperes**

Power and metric prefixes

Power values in electrotechnology can range from gigawatts (GW) to less than a microwatt (μW). For example, coal-fired power stations can produce over 2 GW (2000 megawatts) of power, with each alternator in the power station delivering up to 600 MW. An electric passenger train could need more than 2 MW of power to pull out of the station. The sound system at an outdoor pop concert can produce some 50 kW (kilowatts) of sound, needing about 100 kW of power to drive the system.

On a much smaller scale, iPod or mobile phone earphones take only a few milliwatts (mW) of power for a satisfying sound level. The liquid crystal display in a calculator takes only a few microwatts of power from the battery.

Therefore, as with current, voltage and resistance values, prefixes and engineering notation are often used with power values (see Table 4.1).

POWER WITH METRIC PREFIXES**TABLE 4.1**

Term	Symbol	Equals	Engineering notation
gigawatt	GW	1000 000 000	$1 \text{ GW} = 1 \times 10^9 \text{ W}$
megawatt	MW	1 000 000 watts	$1 \text{ MW} = 1 \times 10^6 \text{ W}$
kilowatt	kW	1000 watts	$1 \text{ kW} = 1 \times 10^3 \text{ W}$
milliwatt	mW	0.001 watts	$1 \text{ mW} = 1 \times 10^{-3} \text{ W}$
microwatt	μW	0.000 001 watts	$1 \mu\text{W} = 1 \times 10^{-6} \text{ W}$

Calculations involving power values expressed with a prefix are treated the same way as Ohm's law equations, as explained in Chapter 3. As a reminder, when multiplying values expressed with engineering notation, you *add* the exponents. For division, the exponent with the term below the line is *subtracted* from the exponent with the term above the line.

EXAMPLE 4.9**FIGURE 4.11**

Solar-powered calculator



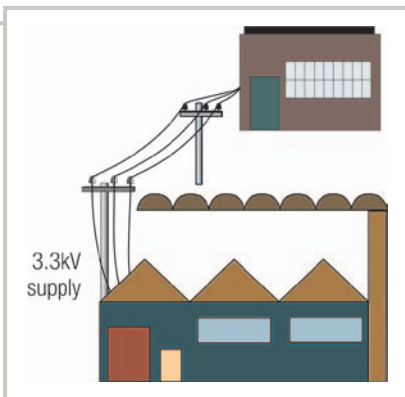
A solar-powered calculator takes $70 \mu\text{A}$ from its solar panel and uses $210 \mu\text{W}$ of power. Find the solar panel voltage.

Solution

Values	$P = 210 \mu\text{W}$
	$I = 70 \mu\text{A}$
	$V = ?$
Equation	$V = \frac{P}{I}$
	$= \frac{210 \times 10^{-6}}{70 \times 10^{-6}}$
Answer	$V = 3 \text{ V}$

EXAMPLE 4.10**FIGURE 4.12**

3.3 kV supply to a factory



A manufacturing company is connected to a 3.3 kV supply and takes 1200 amperes when operating at full production. How much power is being taken from the supply?

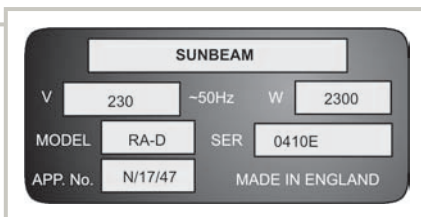
Solution

Values	$V = 3.3 \text{ kilovolts}$
	$I = 1200 \text{ amperes}$
	$P = ?$
Equation	$P = VI$
	$= (3.3 \times 10^3) \times (1.2 \times 10^3)$
Answer	$P = 3.96 \times 10^6 \text{ W or } 3.96 \text{ MW}$

Power and energy rating

FIGURE 4.13

Electric heater rating plate



Electrical appliances are generally required to be marked to show how much power an appliance will take when connected to its rated voltage. This is the appliance's *power rating*. For instance, as shown in Figure 4.13, a domestic heater might take 2300 watts when connected to 230 V AC. Electric motors usually have a rating plate to show the current taken at full load when the motor is connected to its specified voltage.

Domestic appliances such as clothes driers, refrigerators, air conditioners and dishwashers are required to display an energy consumption label. However, as the label in Figure 4.14 shows, the value is given in kilowatt hours used during normal operation over a year, not joules as you might

have expected. Figure 4.14 is a label for a clothes drier; those on other types of appliances such as air conditioning systems are different.

As stated before, the power is one watt if one joule of energy is transformed in one second. Therefore, if the power remains constant at one watt, 3600 joules of energy will be transformed over one hour (as there are 3600 seconds in an hour). An appliance rated at one kilowatt would use 3 600 000 joules each hour, or put another way, it has used one kilowatt hour (kWh) of energy. That is:

$$\text{energy in joules} = \text{power} \times \text{time}$$

or

$$\text{energy in kWh} = \text{kW} \times \text{hours}.$$

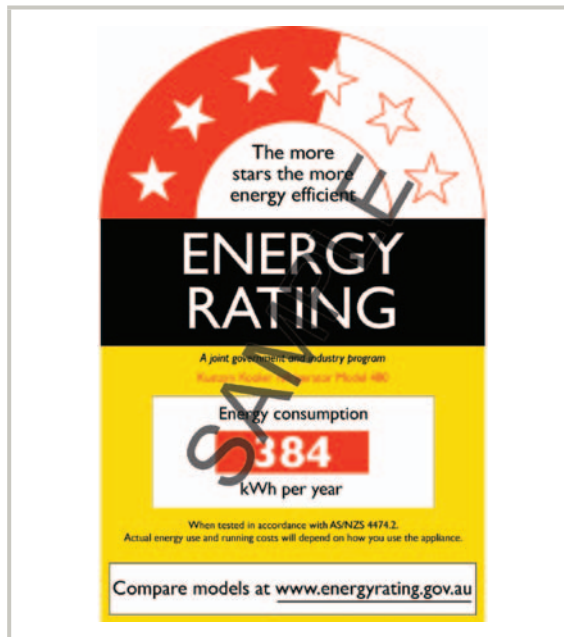


FIGURE 4.14

Appliance energy rating is given in kWh per year

EXAMPLE 4.11

An electric heater rated at 2500 watts is left switched on for two hours. How much energy does it use (a) in joules and (b) in kilowatt hours?

Solution

- (a) Equation energy = power $\times$ time (joules)
 time = $2 \times 60 \times 60 = 7200$ seconds
 power = 2500 watts
 energy = 2500×7200

Answer (a) = **18 000 000 joules or 18 kJ**

- (b) Equation energy = kWh = kW $\times$ hours (kWh)
 time = 2 hours
 energy = 2.5×2

Answer (b) = **5 kWh**



FIGURE 4.15

Electrical energy meter as fitted to many homes. New installations will have a digital energy meter.

Electrical energy is priced by the kilowatt hour. In Example 4.11, if the cost is \$0.18 per kWh, the cost of running this electric heater for two hours is around 90 cents. Electrical energy is measured with a kilowatt-hour meter, which can be either analog or digital. Electrical power is measured with a wattmeter, which can also be analog or digital. A wattmeter measures the circuit current and circuit voltage and outputs a reading in watts. Wattmeters are described in Chapter 25.

SUMMING UP...

- Electrical power (P) is the rate at which electrical energy is transformed.
- Power is measured in watts, where one watt is the power dissipated in a resistance that has one volt across it and one ampere flowing through it.
- Power equals voltage multiplied by current ($P = VI$) watts.
- By transposition, $V = \frac{P}{I}$ and $I = \frac{P}{V}$.
- Electrical power values are often expressed with a metric prefix.
- Electrical energy is measured in kilowatt hours, where $1 \text{ kWh} = 3\,600\,000 \text{ J}$.

EXERCISE 4.2

- 1 Calculate the power taken by an electrical load if the current is 25 A and the voltage across the load is 250 V.
- 2 A 1 kW heating element takes 10 A of current at its correct operating voltage. Calculate the value of the voltage.
- 3 An electric toaster uses 1100 watts of power when operating at 230 V. Calculate the current taken by the toaster.
- 4 The current in a circuit is 25 μA when the voltage is 6 V. Calculate the power consumed by the circuit.
- 5 An industrial plant requires 5 MW of power at a voltage of 11 kV. Calculate the current taken by the plant.

4.5 Power and Ohm's law

Ohm's law gives us a mathematical relationship between current (I), voltage (V) and resistance (R). The three Ohm's law equations introduced in Chapter 3 are:

$$I = \frac{V}{R}$$

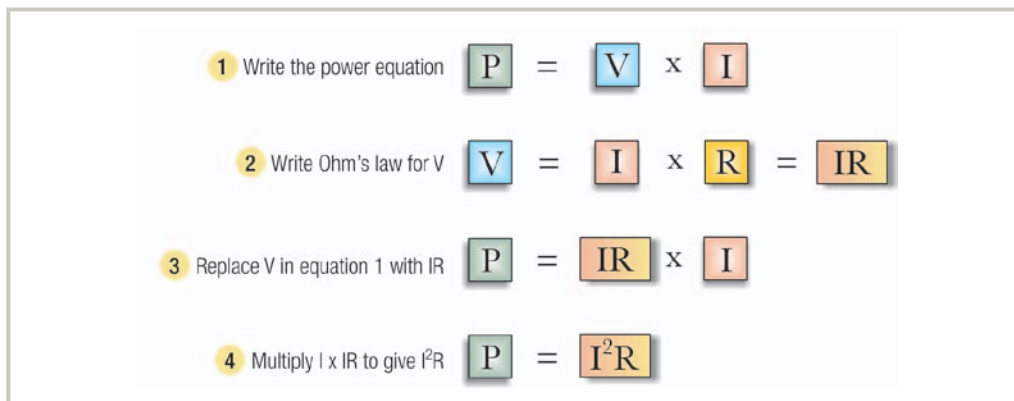
$$V = IR$$

$$R = \frac{V}{I}$$

These and the power equations can be combined to give a set of equations that relate the four quantities of power, voltage, current and resistance. For example, if the resistance and current in a circuit are known, you can determine the voltage using Ohm's law. Then, the equation $P = VI$ can be used to calculate the power dissipated in the circuit. By combining the Ohm's law and power equations, power can instead be directly calculated from the resistance and current values. We start by deriving an equation to find power, if resistance and current are known.

Power from resistance and current

The Ohm's law equation for V can be put in the power equation $P = VI$ to replace the term V. The sequence to do this is shown in Figure 4.16.

**FIGURE 4.16**

Combining Ohm's law and power equations to get power in terms of current and resistance

So, if the current and the resistance are known, the equation to find power is:

$$P = I^2R$$

where:

P = power in watts

I = current in amperes

R = resistance in ohms.

The next two examples use this equation.

EXAMPLE 4.12

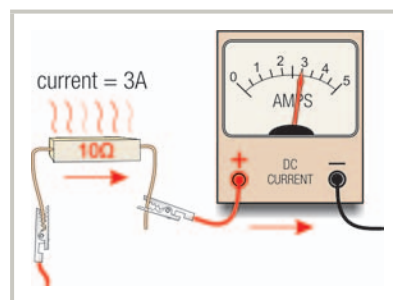
Find the power dissipated in a $10\ \Omega$ resistor when the current through it is 3 A.

Solution

Values $I = 0.3\ \text{A}$
 $R = 10\ \Omega$
 $P = ?$

Equation $P = I^2R$
 $P = 3^2 \times 10$
 $P = 9 \times 10$

Answer **$P = 90\ \text{W}$**

**FIGURE 4.17**

Circuit for Example 4.12

Example 4.13 shows how to use the equation when the values are expressed with metric prefixes.

EXAMPLE 4.13

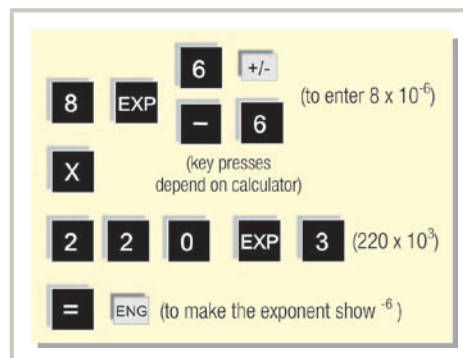
Find the power dissipated by a $220\ \text{k}\Omega$ resistor when the current through it is $8\ \mu\text{A}$.

Solution

Values $I = 8\ \mu\text{A}$
 $R = 220\ \text{k}\Omega$
 $P = ?$

Equation $P = I^2R$
 $= (8 \times 10^{-6})^2 \times (220 \times 10^3)$
 $= (64 \times 10^{-12}) \times (220 \times 10^3)$
 $= (64 \times 220) \times (10^{-12} \times 10^3)$
 $= 14080 \times 10^{-9}$

Answer **$P = 14.08\ \mu\text{W}$**

**FIGURE 4.18**

Calculator key strokes for Example 4.13

Current from power and resistance

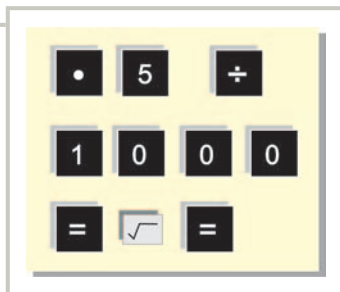
The equation $P = I^2 R$ can be transposed to make either current (I) or resistance (R) the subject. To make I the subject:

- 1 Write the equation: $P = I^2 R$
- 2 Divide both sides by R: $\frac{P}{R} = \frac{I^2 R}{R}$
- 3 Cancel the Rs on the right: $\frac{P}{R} = I^2$
- 4 Take the square root of both sides and rearrange so I is on the left: $I = \sqrt{\frac{P}{R}}$.

EXAMPLE 4.14

FIGURE 4.19

Calculator key strokes for Example 4.14



Find the current flowing in a 1 k Ω resistor if it is dissipating 0.5 W of power.

Solution

Values

$$P = 0.5 \text{ W}$$

$$R = 1000 \Omega$$

$$I = ?$$

Equation

$$I = \sqrt{\frac{P}{R}} = \sqrt{\frac{0.5}{1000}} = \sqrt{\frac{0.5}{10^3}}$$

$$= \sqrt{0.5 \times 10^{-3}}$$

$$= 0.02236 \text{ A}$$

Answer

$$I = \mathbf{22.36 \text{ mA}}$$

Resistance from power and current

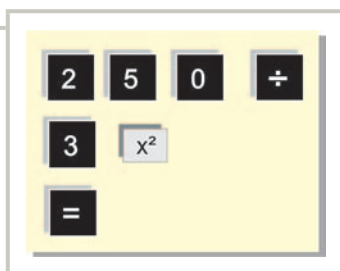
The sequence to make R the subject of $P = I^2 R$ is shown below.

- 1 Write the equation: $P = I^2 R$
- 2 Divide both sides by I^2 : $\frac{P}{I^2} = \frac{I^2 R}{I^2}$
- 3 Cancel the I^2 s on the right and rearrange the equation so R is on the left: $R = \frac{P}{I^2}$.

EXAMPLE 4.15

FIGURE 4.20

Calculator key strokes for Example 4.15



Find the resistance of a heating element that takes 250 W of power at a current of 3 A.

Solution

Values

$$P = 250 \text{ W}$$

$$I = 3 \text{ A}$$

$$R = ?$$

Equation

$$R = \frac{P}{I^2} = \frac{250}{3^2}$$

$$= \frac{250}{9}$$

Answer

$$\mathbf{R = 27.8 \Omega \text{ (rounded off to one decimal place)}}$$

Power from resistance and voltage

To get an equation where power can be found from resistance and voltage values, the Ohm's law equation for current is combined with $P = VI$. The sequence is shown in Figure 4.21.

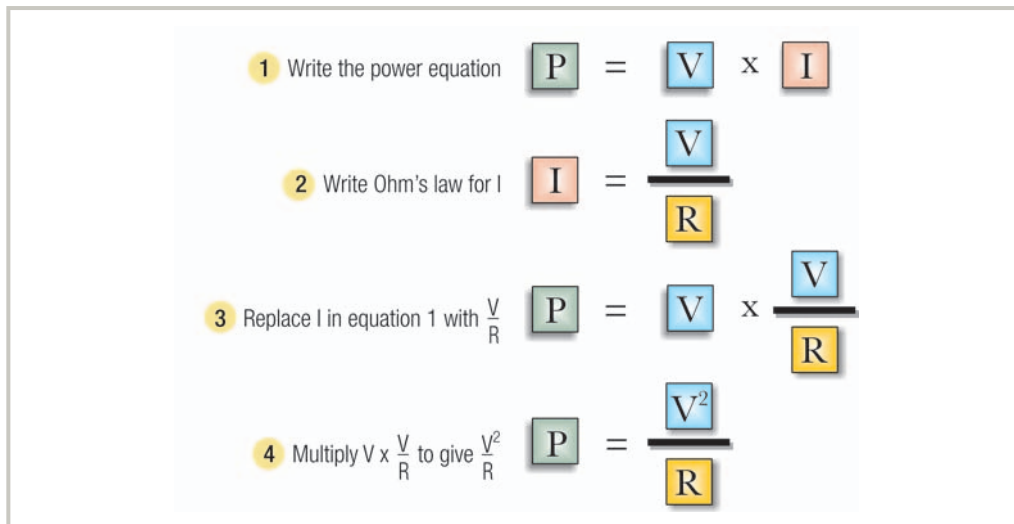


FIGURE 4.21

Sequence to get power in terms of voltage and resistance

So, the equation to find power if the voltage and the resistance are known is:

$$P = \frac{V^2}{R}$$

where:

P = power in watts

V = voltage in volts

R = resistance in ohms.

EXAMPLE 4.16

Find the power dissipated by a $120\ \Omega$ resistor when the voltage across it is 10 V.

Solution

Values $V = 10\text{ V}$
 $R = 120\ \Omega$
 $P = ?$

Equation $P = \frac{V^2}{R} = \frac{10^2}{120} = \frac{100}{120}$

Answer **$P = 0.833\text{ watts}$**

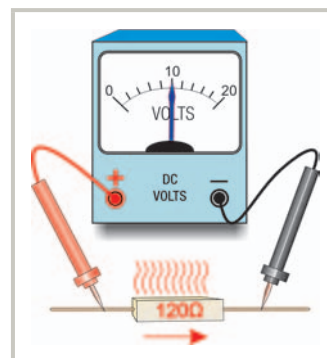


FIGURE 4.22

Circuit for Example 4.16

Resistance from voltage and power

If resistance is the unknown value, make R the subject in the equation, as shown below.

1 Write the equation being transposed: $P = \frac{V^2}{R}$

2 Multiply both sides by R : $P \times R = \frac{V^2}{R} \times R$

3 Cancel the Rs on the right: $PR = V^2$

4 Divide both sides by P: $\frac{PR}{P} = \frac{V^2}{P}$

5 Cancel the Ps: $R = \frac{V^2}{P}$.

So, the equation to find resistance if the voltage and power are known is:

$$R = \frac{V^2}{P}$$

where:

R = resistance in ohms

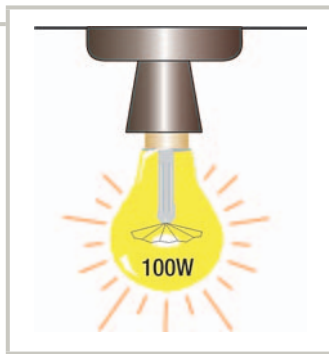
V = voltage in volts

P = power in watts.

EXAMPLE 4.17

FIGURE 4.23

Lamp in Example 4.17



Find the resistance of a 230 V, 100 W incandescent lamp (resistance when the lamp is on).

Solution

Values $V = 230 \text{ V}$
 $P = 100 \text{ W}$
 $R = ?$

Equation $R = \frac{V^2}{P}$

$$= \frac{230^2}{100} = \frac{230 \times 230}{100} = \frac{52900}{100}$$

Answer **$R = 529 \Omega$**

This value is the resistance of the lamp filament at its operating temperature. If you measure the filament's resistance when it's cold, the value will be much lower, at around 50Ω .

Voltage from power and resistance

If voltage is the unknown value, make V the subject in the equation, as shown below.

1 Write the equation being transposed: $P = \frac{V^2}{R}$

2 Multiply both sides by R: $P \times R = \frac{V^2}{R} \times R$

3 Cancel the Rs on the right: $PR = V^2$

4 Take the square root of both sides and write equation so V is on the left: $V = \sqrt{PR}$.

So, the equation to find voltage if the resistance and power are known is:

$$V = \sqrt{PR}$$

where:

R = resistance in ohms

V = voltage in volts

P = power in watts.

EXAMPLE 4.18

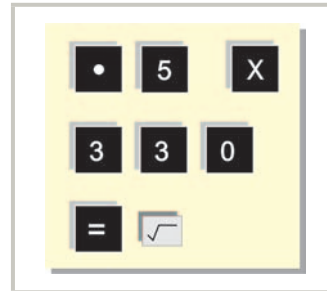
What is the maximum voltage that can be applied across a $330\ \Omega$ resistor that has a maximum power rating of $0.5\ \text{W}$?

Solution

Values $R = 330\ \text{ohms}$
 $P = 0.5\ \text{watts}$
 $V = ?$

Equation $V = \sqrt{PR}$
 $= \sqrt{0.5 \times 330} = \sqrt{165}$

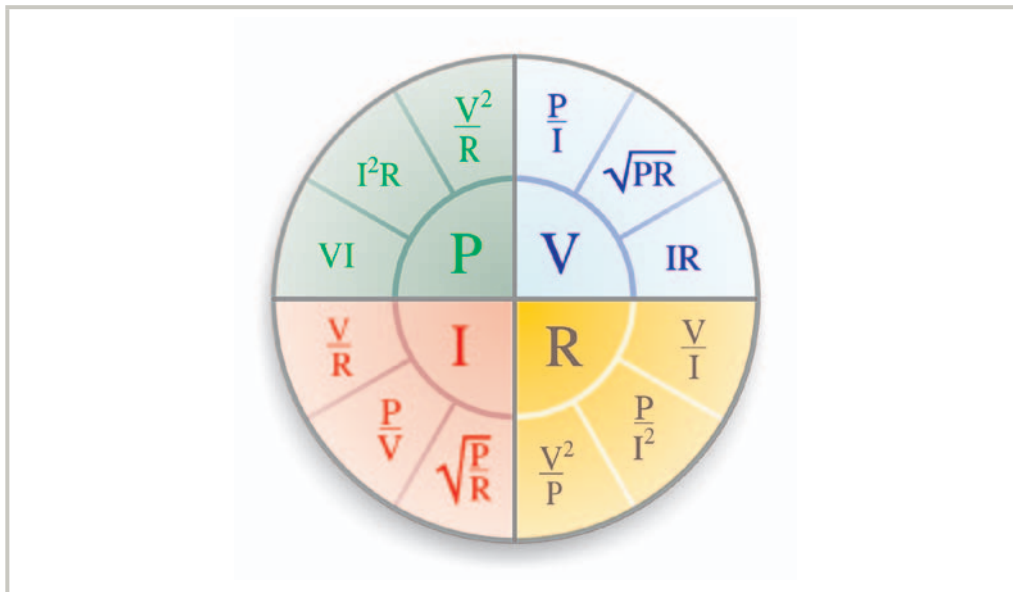
Answer **$V = 12.85\ \text{volts}$**

**FIGURE 4.24**

Calculator key strokes for Example 4.18

SUMMING UP...

- Combining Ohm's law and the power equation $P = VI$ gives nine equations that relate power, voltage, current and resistance.
- If any two of these four quantities are known, the unknown quantities can be calculated.
- Figure 4.25 summarises the nine power equations and the three Ohm's law equations.

**FIGURE 4.25**

Equation wheel. The four quantities V , I , R and P are in the centre section near their three related equations, giving 12 equations. Remember some of the equations, and derive the others as you need them.

EXERCISE 4.3

- Find the power consumed by a circuit that has a resistance of $25\ \Omega$ and is taking a current of $5\ \text{A}$.
- How much current does a circuit with a resistance of $500\ \Omega$ take if it is dissipating $3\ \text{W}$ of power?
- The heating element in a commercial espresso coffee machine is rated at $5.75\ \text{kW}$ and takes a current of $20\ \text{A}$. Find the resistance of the element.
- How much power is being dissipated by a $30\ \Omega$ resistor when the voltage across it is $12\ \text{V}$?
- A heater takes $1500\ \text{W}$ of power at its rated voltage of $230\ \text{V}$. What is the resistance of the heater?
- Calculate the voltage across a $25\ \Omega$ load that is dissipating $1\ \text{kW}$ of power.

4.6 Power change with I, V or R changes

If the voltage to a circuit is increased, the current increases proportionally (assuming the resistance stays the same). For example, if the voltage to a circuit is doubled, by Ohm's law the current in the circuit will also double. Because power is the product of voltage and current ($P = VI$), if both voltage and current have doubled in value, the power will be increased by four times. If the resistance of a circuit changes, there will also be a change in the power, but by how much? First a look at what happens when the voltage changes . . .

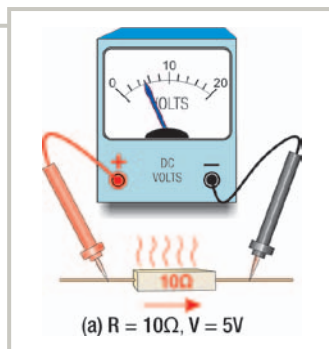
Effect on power if the voltage changes

As explained earlier, the equation relating power, voltage and resistance is $P = \frac{V^2}{R}$. Notice that the equation shows voltage *squared*. This means that the power taken by the circuit will increase by the *square* of the voltage change (assuming the resistance doesn't change). For instance, as outlined above, if the voltage doubles, the power will increase by four times, as two squared (2^2) equals 4. The following example shows this.

EXAMPLE 4.19

FIGURE 4.26

Circuit for
Example 4.19(a)



How much power does a $10\ \Omega$ resistor take when the voltage across it is (a) 5 V, (b) 10 V?

Solution

Values

$$V = 5\text{ V}$$

$$R = 10\ \Omega$$

$$P = ?$$

Equation

$$P = \frac{V^2}{R} = \frac{5^2}{10} = \frac{25}{10}$$

(a) Answer **$P = 2.5\text{ W}$** (when voltage is 5 V)

When the voltage doubles to 10 V, the power is found the same way, as shown below.

Values

$$V = 10\text{ V}$$

$$R = 10\ \Omega$$

$$P = ?$$

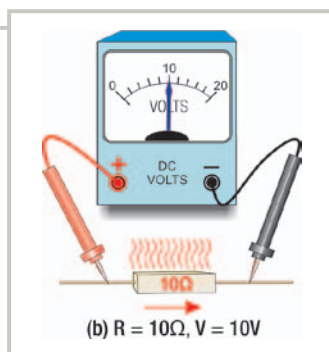
Equation

$$P = \frac{V^2}{R} = \frac{10^2}{10} = \frac{100}{10}$$

(b) Answer **$P = 10\text{ W}$** (when voltage is 10 V)

FIGURE 4.27

Circuit for
Example 4.19(b)



As you can see, the power has increased by four times (as $2.5\text{ W} \times 4$ equals 10 W). Because power changes by the *square* of the voltage, if the voltage increases by three times, the power taken by the resistor will increase by nine times. If the voltage falls by half, the power will fall to a quarter of its previous value. So, a small change in the voltage causes a much larger change in the power taken by a circuit.

Effect on power if the current changes

The equation $P = I^2R$ shows that the power changes with the *square* of the current (providing the resistance doesn't change). That is, if the current through a circuit doubles (while its resistance remains the same), the power dissipated by the circuit increases by four times. The following example shows this effect.

EXAMPLE 4.20

A 25 m extension lead has a total resistance of $2\ \Omega$. Calculate the power dissipated over the length of the lead when (a) the current in the lead is 5 A and (b) when the current is 10 A.

Solution

Values	$I =$ (a) 5 A (b) 10 A
	$R = 2\ \Omega$
	$P = ?$
Equation	$P = I^2R$
(a)	$= 5^2 \times 2$
	$= 25 \times 2$
Answer	(a) $P = 50\ \text{W}$ (when current is 5 A)
(b)	$P = I^2R$
	$= 10^2 \times 2$
	$= 100 \times 2$
Answer	(b) $P = 200\ \text{W}$ (when current is 10 A)



FIGURE 4.28

Extension lead in Example 4.20

Example 4.20 shows that the power dissipated by the lead has increased fourfold from 50 W to 200 W when the current in the lead is doubled. The power is dissipated as heat energy, which is a loss, sometimes referred to as an 'I squared R loss' (I^2R). If the extension lead in Example 4.20 is coiled, the heat generated by the power loss at 10 A could cause it to overheat. If the current in an overhead power line exceeds a certain limit, the heat generated in the cable could cause it to sag and come into contact with other cables.

A bad connection can cause heat to be generated at the connection because of the current flowing at a higher than normal resistance. The heat being produced will cause oxidation and other effects that make the connection's resistance rise even further, in turn producing more heat. Eventually it will fail, and possibly cause heat damage to surrounding areas or start a fire.



Bad electrical connections, particularly in roof cavities, are responsible for numerous house fires. Carefully check all connections you make.

Effect on power if resistance changes

If the voltage to a circuit remains the same and the circuit resistance falls by half, Ohm's law tells us that the current will double. The power to the circuit will also double, as $P = VI$, and the current has increased by two, while the voltage has remained the same. Therefore, the power changes *proportionally* to a change in resistance.

An example is a household electric radiator that has two or three different size heating elements, selected by a switch on the radiator. On the low setting, the switch will select the element with the highest resistance, as it will produce the least amount of heat. To obtain twice the heat, an element with half the resistance of the previous element is selected instead.

Because each element will be connected in turn to the same voltage, we can say that if the voltage remains the same, *power dissipation is inversely proportional to the resistance*. That is, the higher the resistance, the less power dissipated for the same voltage.

CHAPTER SUMMARY

- Energy and work are both measured in joules. Energy is the ability to do work, and work is done when energy is transformed.
- Force is measured in newtons (N). One newton is the force needed to accelerate a one kilogram mass by a rate of one metre per second per second.
- Motors are rated by their power and torque, where torque is measured in newton metres (Nm) and power is measured in watts.
- Energy is either kinetic (moving) or potential (stored). Potential energy becomes kinetic energy when used.
- Power is the rate of doing work. That is:

$$\text{power} = \frac{\text{work}}{\text{time}}$$
 where power is in watts, work is in joules and time is in seconds. The power is one watt if one joule of energy is transformed in one second.
- Mechanical power of a motor: $P = \frac{2\pi NT}{60}$ watts, where N = RPM, T = torque (Nm).
- Efficiency: $\eta = \frac{\text{power out}}{\text{power in}} \times 100\%$. Efficiency gives a measure of power and energy losses.
- Electrical power (P) to a load equals the product of the current in the load and the voltage across the load ($P = VI$).
- $P = VI$ is transposed to make either V or I the subject, and is combined with Ohm's law to give nine equations relating to circuit power, resistance, current and voltage. (See Figure 4.25.)
- $P = \frac{V^2}{R}$ means power changes by the square of the voltage change.
- It also means that power changes inversely with resistance (when the voltage is constant).
- $P = I^2R$ means power changes by the square of the current change.
- 'I squared R losses' refer to the losses generated as heat in a conductor or connection.

SELF-CHECK

We recommend doing the exercises first. Then answer these questions and check your answers at the back of the book.

- 1 Define the term 'energy'.
- 2 What is the relationship between energy and work?
- 3 What is the definition of one newton?
- 4 What is meant by 'one joule of work'?
- 5 When tightening a bolt with a spanner, how much torque is produced when you apply a 300 N force 150 mm from the centre of rotation?
- 6 What is the difference between kinetic energy and potential energy? Give two examples of each type that are used to generate electricity.
- 7 Define the term 'power' and give the general equation for power.
- 8 Give the equation to find the mechanical power of a motor.
- 9 A motor takes 750 W of electrical power and delivers 630 W of mechanical power. Calculate the efficiency of the motor.
- 10 Find the current flowing in a 1.5 kW heater element connected to 120 V.
- 11 If the heater element in Question 10 is connected to a 240 V supply, how much power will it be dissipating (approximately)?
- 12 A commercial oven is rated at 400 V, and has three elements that give a combined resistance (when operating) of 16 ohm. Calculate the power.
- 13 If the oven in Question 12 runs for 10 hours, how much electrical energy (in kWh) has it used? Express the answer with engineering notation.
- 14 For the circuit in Figure 4.29 on the next page, find the resistance (R) and the power (P) dissipated in the resistance for the following meter readings:
 - a ammeter shows 3.5 A, voltmeter shows 230 V
 - b ammeter shows 6 μ A, voltmeter shows 24 V
 - c ammeter shows 23 mA, voltmeter reads 400 V

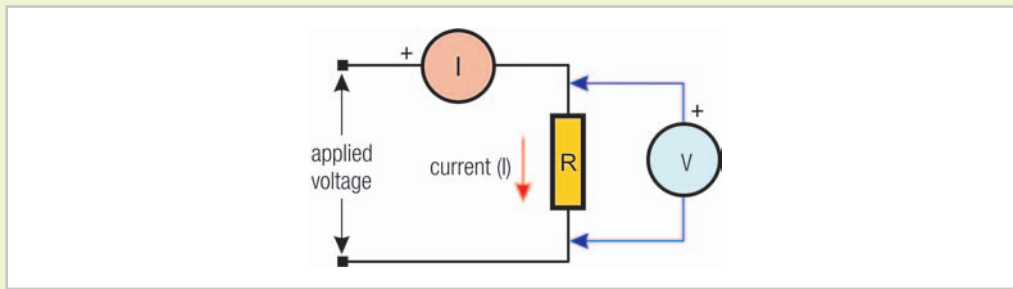


FIGURE 4.29

- 15 For the circuit in Figure 4.30 below, find:
- a Resistance if the applied voltage is 230 V
 - b Current if the resistance is 24 ohm
 - c Voltage if the resistance is 250 ohm

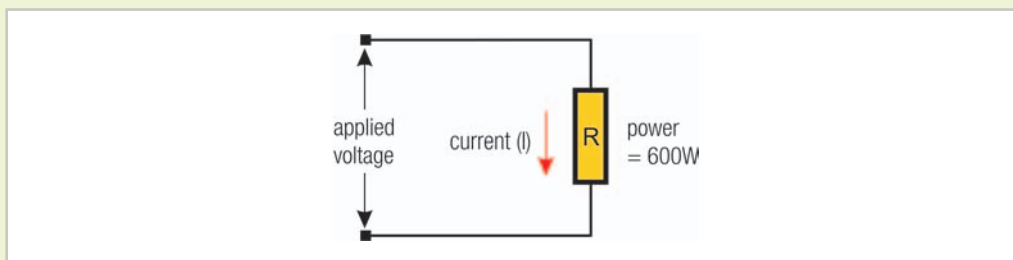


FIGURE 4.30

- 16 A circuit connection has a resistance of 0.15 ohm. How much power is dissipated as heat in the connection when the current through it is 100 A?
- 17 An electric stove connected to a 230 V supply has three hot plates, two rated at 900 W and a larger one rated at 1500 W. Calculate the current taken by the stove when:
- a One 900 W hot plate is operating
 - b Both 900 W hot plates are operating
 - c All hot plates are operating
- 18 Calculate the resistance of the stove for each of the conditions in Question 17.
- 19 A switch is dissipating 45 W of power as heat when it is passing a current of 15 A. Calculate the resistance of the switch.
- 20 After cleaning the contacts of the switch in Question 19, its resistance now measures $500 \mu\Omega$. Calculate the power being dissipated when the current is 15 A.

ONLINE RESOURCES

For more information about the contents of this chapter, visit the following websites:

- www.physicsclassroom.com (click on The Physics Classroom Tutorial for a list of topics to select from, such as Work, Energy and Power)
- www.howstuffworks.com (click on Science, then Physical Science and select from a list of topics, such as electricity or mechanics).